

Wireless Access Electromagnetics

Assignment 1

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1.1 Numerical Solution

First, I computed the reflection coefficients at each port, followed by determining the relation between k and l , from which s_{11} is derived as a function of f/f_0 .

Problem #01

$$Z_{c1} = \sqrt{Z_0 \times Z_{l1}} = 70.71$$

$$Z_{c2} = \sqrt{Z_0 \times Z_{l2}} = 100$$

$$\Gamma_{B1}^- = \frac{Z_{l1} - Z_{c1}}{Z_{l1} + Z_{c1}} = 0.17$$

$$\Gamma_{B2}^- = \frac{Z_{l2} - Z_{c2}}{Z_{l2} + Z_{c2}} = 0.33$$

$$\Gamma_{A1}^+ = \Gamma_{B1}^- e^{-2\pi j k l} = \Gamma_{B1}^- e^{-j \pi f/f_0}$$

$$k \cdot l = \frac{2\pi}{\lambda g} \times \frac{1}{4} = \frac{\pi f}{2f_0}$$

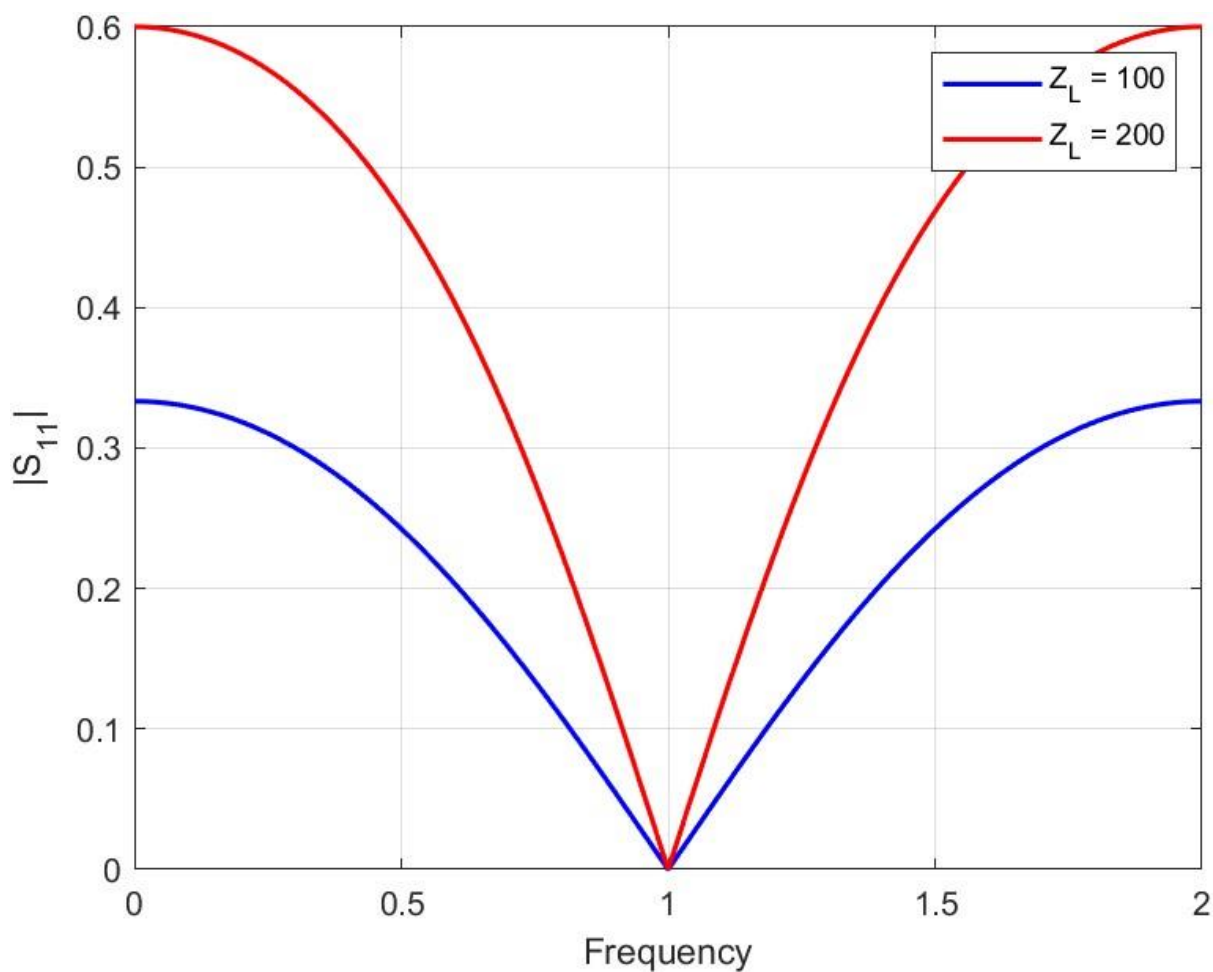
$$Z_{A1} = Z_{c1} \frac{1 + \Gamma_{A1}^+}{1 - \Gamma_{A1}^-}$$

$$Z_{A2} = Z_{c2} \frac{1 + \Gamma_{A2}^+}{1 - \Gamma_{A2}^-}$$

$$\Gamma_{A1}^- = \frac{Z_{A1} - Z_0}{Z_{A1} + Z_0} = S_{11}$$

$$\Gamma_{A2}^- = \frac{Z_{A2} - Z_0}{Z_{A2} + Z_0} = S_{11}$$

1.2. Frequency behaviour of S11



Discussion:

S11 moves around as the frequency changes, which is what we expect from a frequency-dependent matching network.

The $Z_L = 200$ curve sits much higher than $Z_L = 100$ throughout, as it proves that the bigger the load impedance, the harder it is to match.

Both curves hit zero at $f/f_0 = 1$, showing a perfect matching between both the impedances.

Exercise 2:

2.1 Numerical Solution

Pb #2,

$$S_{21} = \frac{b_2}{a_1} \Big|_{a_2=0} = \frac{V_{B+}^+}{V_{A-}^+} \Big|_{V_{B+}^- = 0}$$

$$V_{A+}^+ = (1 + \Gamma_A^+) V_{A-}^+ = V_{A-}^+ (1 + \Gamma_{A-})$$

$$V_{B+}^+ = (1 + \Gamma_B^-) e^{-jkl} \frac{(1 + \Gamma_{A-})}{(1 + \Gamma_{A+})} V_{A-}^+$$

$$V_{A+}^+ = V_{A-}^+ \frac{(1 + \Gamma_{A-})}{(1 + \Gamma_{A+})}$$

$$S_{21} = \frac{(1 + \Gamma_B^-) (1 + \Gamma_{A-}) e^{-jkl}}{(1 + \Gamma_{A+})}$$

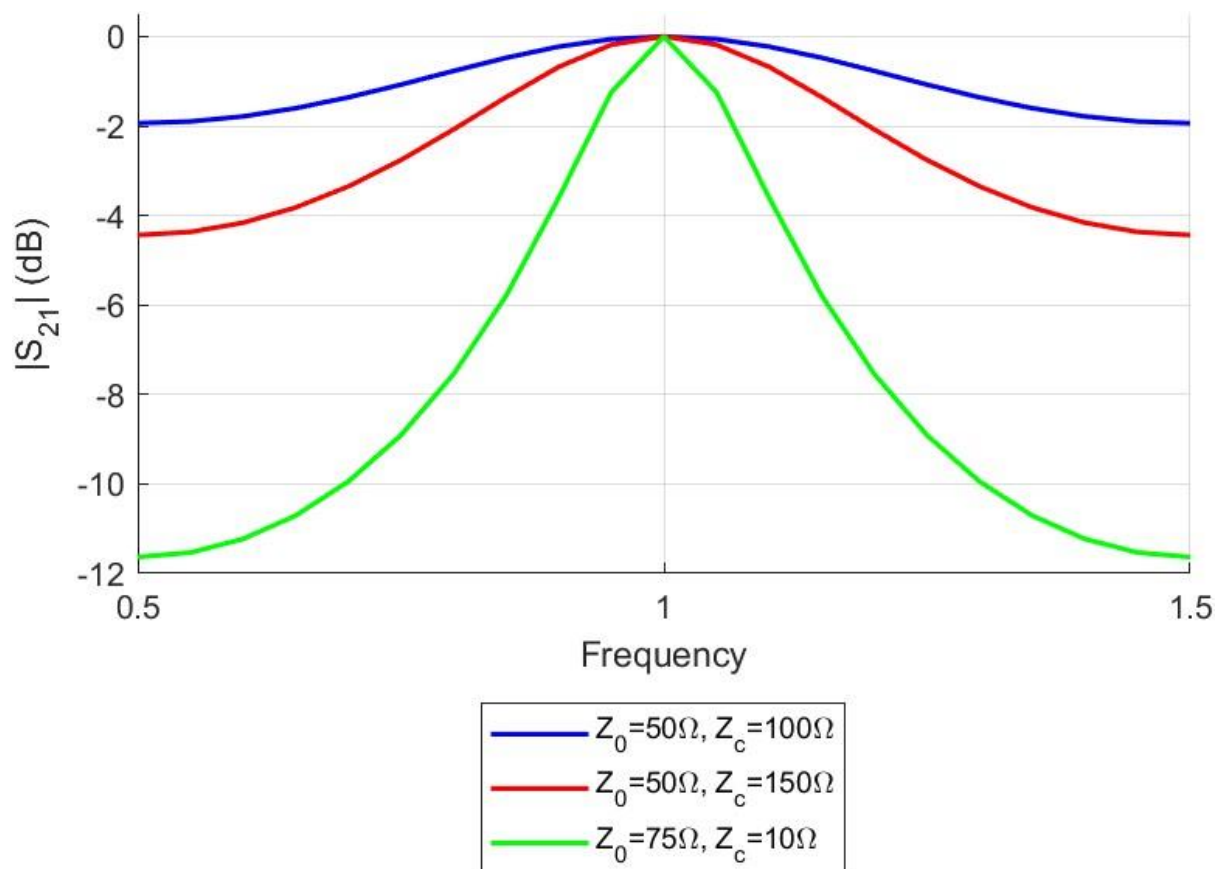
$$\Gamma_A^- = \frac{Z_A - Z_0}{Z_A + Z_0} \quad Z_A = Z_0 \frac{1 + \Gamma_{A+}}{1 - \Gamma_{A+}}$$

$$\Gamma_{A+} = e^{-j2kl} \Gamma_B^-$$

$$\Gamma_B^- = \frac{Z_0 - Z_c}{Z_0 + Z_c} \quad \Gamma_{B1}^- = -0.33 \quad \Gamma_{B2}^- = -0.5 \quad \Gamma_{B3}^- = 0.767$$

2.2 Frequency behaviour of S21

By running the Matlab script, I computed and plotted the scattering parameters as a function of f/f_0 in dB scale.



Discussion:

All three cases reach 0 dB at the design frequency f_0 , meaning full transmission occurs at that point.

The wider the impedance mismatch between Z_0 and Z_c , the narrower and steeper the response curve becomes.

The green curve drops the most sharply, showing that a large mismatch leads to significant signal loss away from f_0 .

It can be demonstrated that s_{21} reaches a value of 0 dB at the design frequency